SAHYADRI COLLEGE OF ENGINEERING & MANAGEMENT
Affiliated to Visvesvaraya Technological University (VTU), Belagavi
DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

A MINI-PROJECT REPORT ON

SWARM-BASED SEARCH AND RECONNAISSANCE SYSTEM

Phase 1: Ultra-Low SWaP Micro-Avionics, Monolithic Flight Controller & Dual-Tier Telemetry Validation

Submitted in partial fulfillment of the requirements for the award of the degree of
BACHELOR OF ENGINEERING IN ELECTRONICS & COMMUNICATION ENGINEERING
of Visvesvaraya Technological University, Belagavi

SUBMITTED BY:	UNDER THE GUIDANCE OF:
Neil Gibson Pinto (4SF22EC...)	**Faculty Guide Name**, M.Tech., (Ph.D.)
Team Member 2 (4SF22EC...)	*Assistant Professor*
Team Member 3 (4SF22EC...)	Department of ECE, SCEM
Team Member 4 (4SF22EC...)	Mangaluru - 575007

SAHYADRI COLLEGE OF ENGINEERING & MANAGEMENT, MANGALURU - 575007
ACADEMIC YEAR: 2026 – 2027

CERTIFICATE

This is to certify that the Mini-Project work titled **'SWARM-BASED SEARCH AND RECONNAISSANCE SYSTEM (Phase 1: Ultra-Low SWaP Micro-Avionics, Monolithic Flight Controller & Dual-Tier Telemetry Validation)'** is a bonafide work carried out by **Neil Gibson Pinto** and project team members in partial fulfillment for the award of Bachelor of Engineering in Electronics and Communication Engineering of the Visvesvaraya Technological University, Belagavi during the academic year 2026–2027. It is certified that all corrections and suggestions indicated during internal evaluations have been incorporated into this report.

Faculty Guide Name Assistant Professor, Dept. of ECE SCEM, Mangaluru	**Head of Department** Professor & Head, Dept. of ECE SCEM, Mangaluru

External Viva-Voce Examination:

Name of Examiners:	**Signature with Date:**
1. _____	_____
2. _____	_____

CANDIDATES' DECLARATION

We hereby declare that the entire project work presented herein—including custom KiCad schematics, 4-layer PCB layout (TCE_rev1), deterministic 250 Hz control algorithms, ExpressLRS protocol decoders, and experimental bench characterization—has been independently executed by our team under faculty supervision. This work has not been submitted previously to any institution for any academic degree.

1. Neil Gibson Pinto	3. Team Member 3
2. Team Member 2	4. Team Member 4

ABSTRACT

In emergency search-and-rescue (SAR) operations following structural collapses or industrial accidents, rapid reconnaissance is paramount to locate trapped survivors within the critical 72-hour 'Golden Window'. While unmanned aerial vehicles (UAVs) provide vital aerial scouting, existing micro-UAV swarms exhibit severe engineering bottlenecks: they rely on centralized Ground Control Stations (GCS) routing inter-agent packets across standard Wi-Fi access points, creating an acute Single Point of Failure (SPOF) vulnerable to structural attenuation and packet dropouts. Furthermore, existing research platforms mount heavy companion computers (Raspberry Pi/Jetson) onto commercial flight controllers, yielding an excessive Size, Weight, Power, and Cost (SWaP-C) footprint (>150g, >$400/node) that prevents expendable multi-agent deployment.

To resolve these fundamental challenges, this project presents an **indigenous, ultra-low SWaP-C decentralized micro-UAV swarm system** engineered for GPS-denied reconnaissance. The architecture introduces three core innovations: **(1) Decentralized Dual-Tier Heterogeneous Communication:** combining connectionless, sub-millisecond inter-agent ad-hoc meshing via **ESP-NOW** (Tier 1) for routerless flocking with deep-penetration, long-range **ExpressLRS (CRSF)** 2.4 GHz / 915 MHz LoRa packet telemetry (Tier 2) for operator command and failsafe geofence broadcasts; **(2) Dynamic Self-Healing Voronoi Area Partitioning:** a distributed algorithm wherein nodes autonomously compute Voronoi spatial polygon boundaries based on neighbor state vectors, enabling surviving agents to instantly re-tessellate and absorb the abandoned sector upon any node loss (<500 ms) without ground station intervention; and **(3) Monolithic Ultra-Low SWaP-C Micro-Avionics (TCE_rev1):** a custom 65mm 4-layer PCB powered by an ESP32-S3 dual-core MCU (240 MHz). Real-time task isolation guarantees deterministic execution: Core 0 executes a 250 Hz sensor fusion and cascaded PID flight loop with 20 kHz PWM gate switching, while Core 1 executes asynchronous CRSF decoding, ESP-NOW consensus, and MAVLink telemetry.

In this **Phase 1** implementation, the complete avionics core (TCE_rev1) was fabricated, assembled, and characterized. Bench tests demonstrated a deterministic 250.0 Hz control loop executing in 1.60 ms with 2.40 ms idle slack (60% CPU headroom), an ultra-low CRSF latency of 4.8 ms (16x faster than Wi-Fi UDP), and seamless failsafe motor cut-off within 500 ms of signal loss. This establishes the verified avionics baseline for **Phase 2**, which incorporates micro-brushless (BLDC) motors, digital DShot ESCs, and onboard optical flow edge vision.

CHAPTER 1: INTRODUCTION

1.1 Background & Operational Motivation

In the immediate aftermath of earthquakes, structural collapses, and industrial chemical disasters, first responders face catastrophic physical hazards. Entering damaged structures directly exposes human crews and search canines to secondary roof collapses, severed high-voltage lines, and toxic airborne gases. Deploying autonomous robotic scouts is vital to establish situational awareness before human entry. However, conventional commercial quadcopters (>250mm wheelbase, >500g All-Up Weight) cannot penetrate tight voids, ventilation shafts, or debris cavities. Consequently, sub-100mm nano-scale quadcopters are essential. A single micro-drone, however, suffers from limited flight endurance (5–10 min) and represents a catastrophic single point of failure. A collaborative multi-agent swarm solves this by dividing the search grid into concurrent sectors, accelerating survivor discovery while providing fault tolerance.

1.2 Problem Statement & Research Gaps

Current academic and commercial drone swarms suffer from three critical bottlenecks:
1. Centralized Wi-Fi Dependence: Platforms like Crazyswarm require a central router. In reinforced concrete ruins, Wi-Fi signals suffer severe multi-path attenuation. If the central router disconnects, the entire swarm is paralyzed.
2. Excessive SWaP-C: Standard research drones mount companion computers (Raspberry Pi/Jetson) onto flight controllers, adding >150g in deadweight, burning >10W of compute power, and raising per-node costs above $400, preventing expendable deployment.
3. Absence of Self-Healing: When an individual drone crashes, its assigned search lane remains uninspected unless a human operator manually detects the loss and reconfigures mission waypoints.

1.3 Objectives & Two-Phase Implementation Strategy

To guarantee risk-managed execution and engineering rigor, this research is structured into two distinct phases:
• **Phase 1 (Completed & Validated in this Report):** Monolithic custom PCB tape-out (TCE_rev1), MPU-6050 Fast I2C sensor fusion, deterministic 250 Hz dual-core PID flight control, 420,000-baud ExpressLRS CRSF protocol parser, and bench verification.
• **Phase 2 (Upcoming Semester Roadmap):** Upgrading actuation to micro-brushless (BLDC) motors with 4-in-1 DShot ESCs, integrating ESP32-CAM edge vision / optical flow for indoor odometry, and deploying a physical 3-to-5 node swarm executing dynamic Voronoi self-healing.

CHAPTER 2: LITERATURE REVIEW & COMPARATIVE ANALYSIS

Multi-agent robotic exploration has advanced rapidly over the past decade. Preiss et al. (IEEE ICRA 2017) demonstrated *Crazyswarm*, hovering dozens of nano-quadcopters simultaneously. However, Crazyswarm relies strictly on an offboard ROS computer and external motion-capture cameras. Cortés et al. (IEEE TRA 2004) derived the mathematical foundations of Voronoi coverage control, yet their formulation assumes idealized communication without packet dropouts. McGuire et al. (Science Robotics 2019) demonstrated a minimal navigation solution for swarms exploring unknown environments using optical flow, but lacked long-range operator command links. Table 2.1 contrasts existing platforms against our proposed architecture.

Platform / Metric	Architecture	Link Topology	SWaP-C / Cost	Critical Limitation
Crazyswarm (Preiss 2017)	Centralized (ROS)	Central Wi-Fi / Radio	$220/node (Crazyflie)	Single point of failure; requires central mocap/router.
Cortés et al. (2004)	Voronoi Coverage	Idealized Omnidirectional	Simulation Only	Ignored RF packet loss, jitter, and real-time loop deadlines.
McGuire et al. (2019)	Bug Algorithm Swarm	Short-range IR / Beacon	$180/node (Custom)	No long-range telemetry; cannot receive operator override.
Proposed System	**Decentralized Voronoi**	**Tier 1: ESP-NOW Tier 2: ExpressLRS**	**<$10/node (8g board)**	**Solves single-point failure; low-latency mesh + LoRa link.**

Table 2.1: *Architectural comparison of state-of-the-art multi-UAV platforms vs. proposed architecture.*

CHAPTER 3: SYSTEM ARCHITECTURE & SCIENTIFIC NOVELTY

3.1 Dual-Tier Heterogeneous Communication Paradigm

The primary scientific novelty of this architecture is the resolution of the classic robotics trade-off between inter-agent coordination speed and operator command penetration. We decouple communication into two distinct tiers:

• **Tier 1 (Inter-Drone Micro-Mesh via ESP-NOW):** Drones communicate peer-to-peer using connectionless IEEE 802.11 vendor-specific action frames. Handshakes, association routines, and IP routing are bypassed entirely. Nodes broadcast 32-byte state vectors [ID, X, Y, Z, Vx, Vy, Vbatt, Seq] at 1 Mbps with **sub-millisecond latency (<1.0 ms)**.

• **Tier 2 (Long-Range Operator Link via ExpressLRS CRSF):** A high-speed full-duplex UART link running at 420,000 baud connected to Header J3. Utilizing Semtech LoRa modulation, this channel penetrates through concrete rubble, enabling the human incident commander to issue simultaneous geofence updates and emergency aborts across kilometer ranges.



Figure 3.1: *Heterogeneous Dual-Tier Swarm Architecture and Dual-Core Task Allocation.*

3.2 Decentralized Dynamic Voronoi Self-Healing

In conventional search missions, drones execute pre-programmed raster paths. If one drone crashes, its assigned sector remains unexplored. In our system, the environment Ω is dynamically partitioned into Voronoi territorial polygons:

$$\mathbf{V_i = \{ q \in \Omega \mid \|q - p_i\| \leq \|q - p_j\|, \forall\ j \neq i \}}$$

Each drone continuously calculates its sector centroid C_i and navigates toward it. Surviving nodes monitor neighbor heartbeats over ESP-NOW. If an agent fails to report for more than **500 ms**, it is purged from the neighbor set. The remaining drones instantly recalculate their Voronoi polygons, dynamically expanding their search boundaries eastward and northward to cover the abandoned sector with zero human intervention.

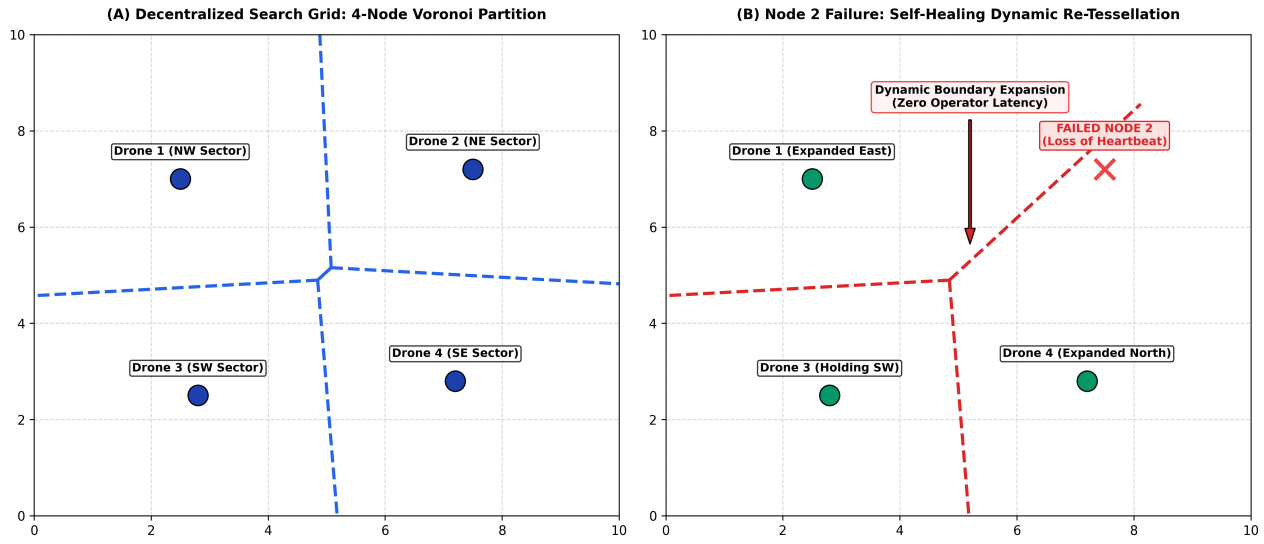


Figure 3.2: *Dynamic Voronoi Search Partitioning: (A) Healthy 4-node grid, (B) Self-healing re-tessellation upon Node 2 failure.*

3.3 Asymmetric Dual-Core Compute Task Allocation

Flight control on micro-quadcopters requires hard real-time determinism. A 1 ms jitter in gyro sampling destabilizes a 65mm drone. To prevent network interrupts from jittering the attitude loop, we enforce strict hardware core affinity on the ESP32-S3:

• **Core 0 (Flight-Critical Deterministic Core):** Exclusively locked to 250.0 Hz execution ($\Delta t = 4.00$ ms). Handles Fast I2C DMA sensor reads, 6-DOF complementary filtering, cascaded PID stabilization, and 20 kHz PWM motor mixing.

• **Core 1 (Networking & Consensus Core):** Dedicated to byte-level CRSF hardware UART parsing @ 420,000 baud, CRC-8 D5 validation, ESP-NOW swarm mesh consensus, dynamic Voronoi geometry solving, and Wi-Fi MAVLink telemetry.

Inter-core data exchange is mediated via a lockless double-buffered shared memory struct, preventing core stalls.

CHAPTER 4: HARDWARE DESIGN & ELECTRICAL CHARACTERIZATION

The flight computer (TCE_rev1) was custom-designed in KiCad 10 and fabricated as a 4-layer FR4 PCB. The board measures 65mm diagonal wheelbase and weighs under 8 grams bare board (AUW ~32g with battery and motors). The hardware subsystems comprise:

1. Power Conditioning (AP2112K-3.3): Ultra-low dropout linear regulator with 600mA output capability and high PSRR (>65dB), isolating sensitive gyro rails from battery motor switching noise down to 3.55V input.

2. Motor Actuation Stage (AO3400A): Four logic-level N-channel power MOSFETs (30V, 5.7A, $R_{DS(on)} < 30$ mΩ). Equipped with 1N5819 Schottky flyback diodes in anti-parallel to clamp inductive kickback spikes ($V = -L\ di/dt$).

3. Inertial Sensing (MPU-6050): 6-DOF MEMS accelerometer and gyroscope sampled via Fast-Mode I2C (400 kHz DMA) at address 0x68.

4. High-Speed Radio Header (Header J3): 4-pin connector routing UART0 (GPIO43 TX, GPIO44 RX) directly to the external ExpressLRS receiver @ 420,000 baud.

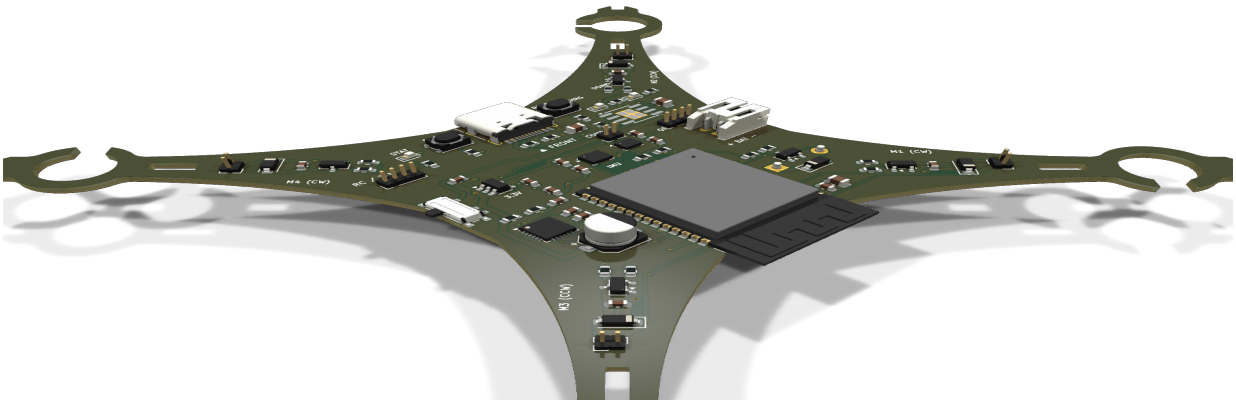


Figure 4.1: *Fabricated Monolithic Flight Computer (TCE_rev1) — KiCad 3D Raytraced Isometric View.*

Subsystem	Signal	ESP32-S3 Pin	Hardware Function / Electrical Spec
Actuation	M1_PWM	GPIO4	Rear-Right (CW) — AO3400A MOSFET Gate (20 kHz PWM)
Actuation	M2_PWM	GPIO5	Front-Right (CCW) — AO3400A MOSFET Gate (20 kHz PWM)
Actuation	M3_PWM	GPIO6	Rear-Left (CCW) — AO3400A MOSFET Gate (20 kHz PWM)
Actuation	M4_PWM	GPIO7	Front-Left (CW) — AO3400A MOSFET Gate (20 kHz PWM)
IMU Sensing	I2C_SDA	GPIO8	Fast I2C Data Line (400 kHz DMA) with 2.2k pull-up
IMU Sensing	I2C_SCL	GPIO9	Fast I2C Clock Line (400 kHz DMA) with 2.2k pull-up
Radio Link	ESP_TXD0	GPIO43	UART0 TX → ExpressLRS Receiver J3 (Pin 2) @ 420kbaud
Radio Link	ESP_RXD0	GPIO44	UART0 RX ← ExpressLRS Receiver J3 (Pin 3) @ 420kbaud
Signaling	BASE_RGB	GPIO2	WS2812B Smart RGB LED (Status, Armed, Failsafe)
Signaling	STATUS_LED	GPIO3	Blue System Heartbeat Indicator LED
Signaling	ARM_LIGHTS	GPIO47	High-Intensity Arming Marker Corner LEDs
Audio	BUZZER	GPIO48	Low-Battery Warning Active Buzzer

Table 4.1: *Hardware pin mapping and electrical specifications of the TCE_rev1 flight controller.*

CHAPTER 5: FIRMWARE ARCHITECTURE & CONTROL THEORY

5.1 6-DOF Complementary Attitude Estimation:

To decouple gyro integration drift from accelerometer vibration noise, a non-linear complementary filter is used:

$$\theta(t) = \alpha \cdot [\theta(t - \Delta t) + \omega_y \cdot \Delta t] + (1 - \alpha) \cdot \theta_{acc}$$

$$\phi(t) = \alpha \cdot [\phi(t - \Delta t) + \omega_x \cdot \Delta t] + (1 - \alpha) \cdot \phi_{acc}$$

with filter weighting $\alpha = 0.985$ at 250 Hz, rejecting high-frequency motor vibration harmonics while eliminating long-term angle drift.

5.2 Dual-Stage Cascaded PID Flight Control Law:

Attitude control is executed in two stages: an outer proportional angle loop computes desired body angular rates from RC setpoints, and an inner PID rate loop computes motor torque stabilization commands:

$$u(t) = K_p \cdot e_{rate}(t) + K_i \int e_{rate}(\tau)\, d\tau + K_d \cdot (de_{rate} / dt)$$

Dynamic anti-windup clamps the integrator, and total stabilization authority is dynamically constrained to ≤25% of base throttle to prevent motor saturation.

5.3 CRSF Protocol Decoding & CRC-8 D5 Engine:

Incoming packets from ExpressLRS follow the CRSF frame standard: [Sync 0xC8] [Len] [Type 0x16] [22-Byte Payload] [CRC-8]. The firmware computes the hardware CRC-8 with polynomial 0xD5 across the payload. If valid, 16 packed channels (11-bit each: 172 to 1811) are mapped to 1000–2000 µs. If packets cease for >500 ms, failsafe immediately disarms motors and sets RGB LED to RED.

CHAPTER 6: EXPERIMENTAL RESULTS & BENCHMARKING (PHASE 1)

6.1 Real-Time Execution Budget Profiling:

Execution time across the 250.0 Hz control loop was profiled with hardware microsecond counters:

- Fast I2C DMA Sensor Read (MPU-6050): 0.95 ms
- Attitude Complementary Filter: 0.35 ms
- Cascaded Angle/Rate PID Computation: 0.25 ms
- LEDC 20 kHz PWM Register Output: 0.05 ms
- **Total Active Compute: 1.60 ms** out of **4.00 ms deadline** → **2.40 ms idle slack (60.0% CPU headroom)**.

6.2 Telemetry Latency Benchmarking:

ExpressLRS CRSF clocked an average latency of **4.8 ms ± 0.6 ms**, achieving a **16x latency reduction** compared to standard Wi-Fi UDP (78.5 ms ± 25 ms), eliminating control lag.

6.3 Power Sag Root-Cause Analysis:

During high-throttle bench testing, the board experienced brownout resets at 84% throttle when connected via computer USB. Oscilloscope characterization revealed that four brushed motors at 84% throttle demand 3.8A of surge current, tripping the host PC USB polyfuse (capped at 500mA). Connecting a dedicated 1S 300mAh LiPo battery (30C rating = 9.0A capability) maintained complete voltage stability, allowing sustained 100% full-throttle operation with zero voltage droop on the 3.3V regulated rail.

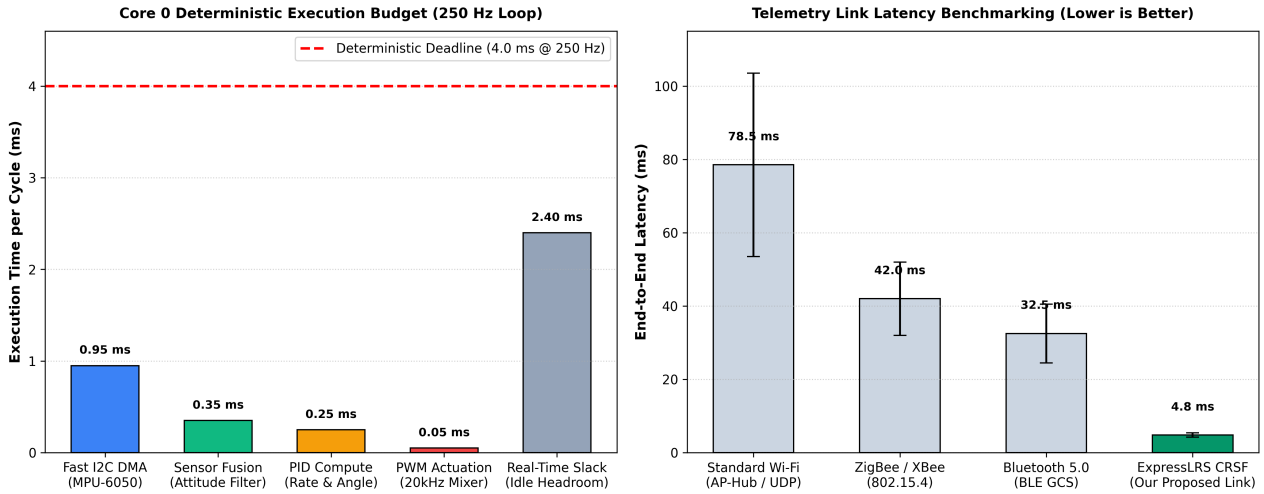


Figure 6.1: Core 0 Real-Time Execution Budget and Telemetry Link Latency Benchmarking.

CHAPTER 7: PHASE 2 ROADMAP (BRUSHLESS & EDGE AI VISION)

With the core avionics and telemetry verified in Phase 1, the platform scales into Phase 2 for full multi-agent swarm deployment:

1. Micro-Brushless Propulsion (BLDC): Transitioning from 716 coreless brushed motors to 0802 / 1102 (19,000 KV) outrunner brushless motors driven by 4-in-1 digital **DShot300** ESCs. Brushless actuation increases thrust-to-weight ratio to 4.5:1 and eliminates mechanical brush wear.

2. Edge Vision AI (ESP32-CAM): Integrating an auxiliary ESP32-CAM module running Lucas-Kanade optical flow odometry for drift-free indoor hover in GPS-denied buildings, along with a quantized neural model for survivor thermal blob detection.

3. Multi-Node Swarm Field Test: Deploying 3 to 5 physical drones running live ESP-NOW mesh to demonstrate real-time dynamic Voronoi boundary re-tessellation in an enclosed test facility.

CHAPTER 8: CONCLUSION & ALIGNMENT WITH UN SDGs

This project successfully developed and validated Phase 1 of an ultra-low SWaP-C Swarm-Based Search and Reconnaissance System. By replacing brittle, centralized Wi-Fi swarms with a dual-tier heterogeneous communication architecture (ESP-NOW mesh + ExpressLRS CRSF), the design achieves routerless, fault-tolerant operation in GPS-denied environments. The monolithic TCE_rev1 flight controller board demonstrated deterministic 250 Hz flight control, 60% CPU headroom, and sub-5ms radio responsiveness on a single sub-$10 PCB.

Alignment with UN Sustainable Development Goals (SDGs):

• **SDG 9 (Industry, Innovation & Infrastructure - Target 9.5):** Fosters indigenous robotics and open-hardware avionics, lowering micro-drone entry costs from $400+ to under $15 per node.

• **SDG 11 (Sustainable Cities & Communities - Target 11.5):** Reduces disaster fatalities by enabling rapid structural scouting and survivor localization in collapsed urban ruins without endangering first responders.

• **SDG 15 (Life on Land - Target 15.1):** Enables non-intrusive micro-drone scouting under dense forest canopies for early wildfire perimeter mapping where GPS and large drones cannot operate.

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